ECE 251: Computer Architecture

Week 12: Exceptions, Interrupts, and Architecture Synthesis

Prof Rob Marano

Spring 2026

1. Chapter 4 Retrospective: Architecture Evolution

- **Single-Cycle**: $CPI = 1.0$, bottlenecked incredibly by the absolute longest instruction format ($T_c \approx 800 - 1000$ps).
- **Multi-Cycle**: FSM driven, faster clock boundaries ($T_c \approx 200$ps), but inflates cycle footprint ($CPI \approx 3 - 5$) operating sequentially.
- **Pipelining**: Overlaps instructions concurrently across D-Flip-Flops to marry the rapid T_c of the Multicycle with the optimal $CPI = 1.0$ limit natively.

2. The Cost of Pipelining: Complexity and Faults

- Pipelining generates extreme throughput at the cost of structural convolution.
- **Data Dependencies**: Mandates massive Forwarding loops to bypass Register caching limits.
- **Control Penalties**: Requires hardware flushing of fetching phases on mispredicted jumps natively.
- Overlapping matrices make stopping instruction tracking incredibly complex when localized sequential constraints fail unexpectedly.

3. Overview: Handling the Unexpected

- Modern processors must systematically handle logic faults or external demands natively without permanently locking up computation.
- **Exceptions**: Synchronous electrical faults generated internally by the datapath executing instructions (e.g., Undefined Opcodes, Math Overflow).
- **Interrupts**: Asynchronous external electrical signals generated outside the datapath execution flow (e.g., Network Packets, I/O Hardware).
- Both conditions force the datapath to immediately halt instruction overlap, backup tracking logic, and jump execution explicitly to an OS handler vector.

4. The Trapping Mechanisms

- **1. Exception Program Counter (EPC)**: A dedicated 32-bit hardware register tasked strictly with capturing the Program Counter memory pointer (PC) of the exact instruction generating the localized exception fault.
- **2. Cause Register**: Records a strict numerical identity classifying exactly *why* the CPU faulted natively, letting the OS run analytical branching routines.
- **3. Vector Handler Routing**: The hardware forces the multiplexer riding the PC tracking pin to dynamically rewire directly to the OS boundary at `0x8000 0180`.

5. SystemVerilog Physics: EPC and Routing

```
// The EPC Latch inside the datapath
always_ff @(posedge clk) begin
  if (Exception_Flag)
    EPC <= PC_Plus4_E - 32'd4; // Capture the failing instruction
end

// The NextPC Multiplexer expansion
always_comb begin
  if (Exception_Flag) NextPC = 32'h8000_0180; // Trap to OS
  else if (PCSrcE)    NextPC = PCBranchE;     // Standard branch
  else                NextPC = PCPlus4F;      // Sequential logic
end
```

6. Pipeline Considerations: Flushing Exceptions

- The biggest challenge is mapping faults correctly across multiple overlapping instructions.
- If an add in the Execute (EX) stage mathematically overflows, the two innocent instructions trailing directly behind it inside the Fetch (IF) and Decode (ID) stages must die instantly.
- The logic strictly utilizes the exact same `StallF` and `FlushD/FlushE` pins engineered for Branch hazards, rewriting the trailing sequence overlapping buffers into inert `NOP` structural states.

eginframe7. Complex Exception Theory: Restartable Precise Exceptions eginitemize

extbfPrecise Exceptions: The MIPS pipeline explicitly guarantees that the state of the processor is saved precisely at the faulting instruction natively. All preceding instructions must cleanly commit, and instruction following it are entirely purged.

extbfRestartable Datapaths: Because we physically track execution limits backwards via the extttEPC (P_C - 4), the OS possesses the native mathematical capacity to transparently re-inject the failed program directly into the fetching queues exactly where it stopped.

extbfImprecise Exceptions: Heavily clustered Out-Of-Order CPU architectures mathematically drop pinpoint recovery limitations, isolating faults to a general "vicinity" and drastically ballooning OS handler complexity constraints natively.
eginframe8. Complex Exception Theory: Multiple Fault Constraints eginitemize

What natively happens if extbftwo isolated instructions concurrently crash across uniquely decoupled pipeline stages during the exact same T_c clock window? (e.g., A memory fault physically intersecting an arithmetic overflow).

extbfHardware Prioritization: The digital architecture intrinsically favors the extitoldest structural instruction actively running (residing mathematically furthest down the pipeline

tracks, such as EX/MEM bounds).

Younger instructions trailing backwards across the IF/ID registers simply haven't formally completed their theoretical lifetimes yet, and thus structurally yield their active interrupts until re-execution.
eginframe[fragile]9. Global Sandbox: extttpipelined_cpu_exceptions eginitemize

All the digital concepts actively synthesized are tracked continuously inside our live extttpipelined_cpu_exceptions hardware namespace.

The entire Patterson & Hennessy theoretical framework natively compiles into executable boolean topologies:
eginverbatim datapath.sv Maps NextPC -¿
$0x8000_0180 OSboundaryhazard.svMapsasynchronousException_F lag->$
$flushloopsmem.svBootstraps.orgboundaryarrayscleanlycontroller.svNativelytracksExceptionbounds$

eginframe10. Debugging Resolutions: 4-Bit ALU Decoupling eginitemize

Incorporating complex math vectors natively clashed with native MIPS bounds.

By natively widening extttalucontrol variables from structural 3-Bit exttt[2:0] to deep 4-Bit

exttt[3:0] limits, we decoupled math logic fundamentally:

Combinational overlap failures mathematically disappeared because Arithmetic Logic (‘4’b0111‘) permanently diverged dynamically away from Sequential structural footprints actively isolating extttMULT (‘4’b1001‘) and extttDIV (‘4’b1000‘) execution bounds!
eginframe11. Debugging Resolutions: Memory Depth & Aliasing eginitemize

Exceptions mechanically redirect hardware to absolute vector limit exttt0x8000 0180.

A generic Datapath limiting array depth to parameter $r = 6$ natively forces Index exttt96 (exttt0x180) to roll back via modulus limits dynamically onto integer exttt32 (exttt0x080).

Expanding system boundaries natively to $r = 8$ establishes a valid OS memory isolation space stretching cleanly to $2^8 = 256$ deep memory frames, granting the OS handler dedicated non-aliasing boundaries natively.
eginframe12. Verification Matrix: The 7 Assembly Playbooks eginitemize

We fully assert the architecture logic leveraging extensive workload matrices simulating MIPS bounds: eginenumerate

extttprog1_simple: Linear testing ensuring Forwarding pipelines.

extttprog2_leaf: Validating Control branches flushing predictions cleanly.

extttprog3_nested: Nested variables generating Load/Use stalls on stacks.

extbf extttprog4_interrupts: Structurally overriding datapath sequences internally dumping active limits securely into kernel space.

13. Retrospective: The von Neumann Architecture Journey

- **Assembly (ISA)**: Human logic bounds mapped natively to binary combinations.
- **Datapath Construction**: ALU calculation grids mapping mathematically mapped alongside immense sequential registry stores.
- **Pipelining Overlaps**: Expanding Single-Cycle bottlenecks geometrically overlapping hardware to maximize processor frequency constraints whilst preserving idealized $CPI = 1.0$.
- **SystemVerilog HDL Control**: SystemVerilog formally proves and describes these complex sequential networks to synthesize logic networks matching photolithography silicon fabrication models reliably.

14. Simulating the Processor (Part 1)

- To functionally test the SystemVerilog components, we must generate a binary executable payload for the CPU's memory arrays.
- **Step 1: Compile Assembly** – Translate your MIPS instructions (`.asm`) down to machine code (`.exe`):

```
$ python3 assembler.py test_prog.asm test_prog.exe
```

- **Step 2: Synthesize Netlist** – Bundle your CPU modules and testbench into an *Icarus Verilog* executable runtime:

```
$ iverilog -g2012 -o pipelined_cpu.vvp tb_computer.sv
```

15. Simulating the Processor (Part 2)

- **Step 3: Run the Simulation** – Execute the compiled runtime. Pass your customized binary into the CPU memory via the runtime `+PROG` hook:

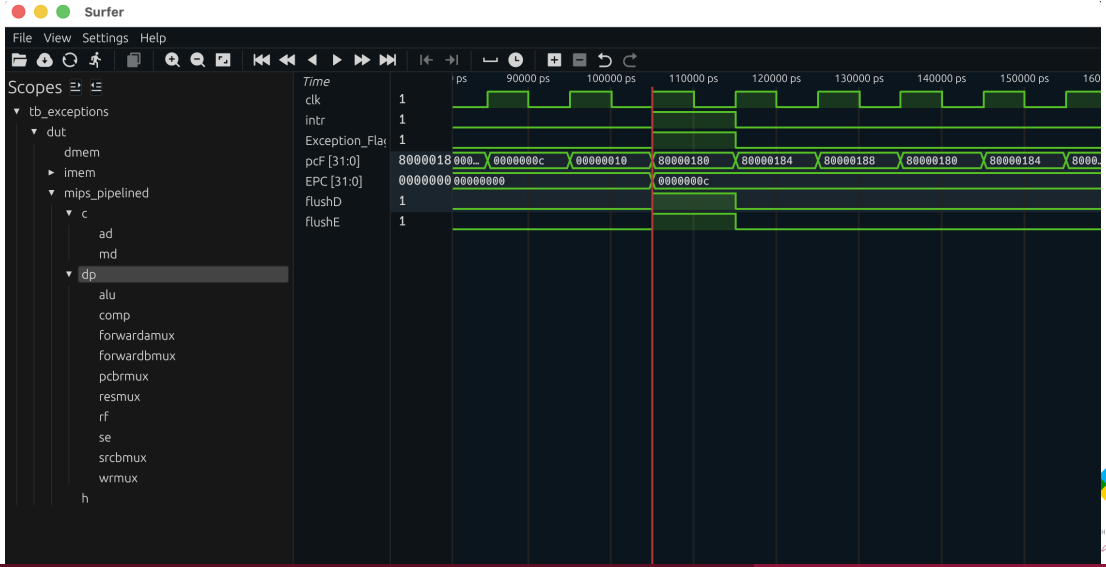
```
$ vvp pipelined_cpu.vvp +PROG=test_prog.exe
```

- The console outputs arithmetic validation metrics confirming Forwarding and Flushing mechanics operated gracefully.
- The testbench drops a digital execution wave file inside your local directory: `tb_computer.vcd`.

16. Reviewing Execution Waves with VS Code "Surfer"

- **Step 4: Inspect the Digital Waveform** – Visually inspect the pipeline delays natively utilizing the VS Code **Surfer** extension!
- Open the generated `tb_computer.vcd` waveform file directly inside Visual Studio Code.
- Track datapath variables across the timeline, heavily inspecting the `EPC`, `FlushD/E`, `StallF/D`, and dynamic ALU Forwarding boundaries.
- **Action Required:** Take meticulous screenshots of your custom Exception faulting waveforms structurally validating logic behaviors within Surfer!

17. Finding the Exception Event in the Waveform



18. Conclusion of the Processor Component

- You now possess the complete hardware logic sequence validating how generic silicon evaluates and processes procedural logic.
- Next steps: Branching beyond the processor logic specifically into Memory Hierarchies and spatial/temporal Caching optimizations spanning across RAM boundaries!